

THE UNIVERSITY OF DODOMA



COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION(CIVE)

COURSE NAME: SELECTED TOPICS IN CYBER SECURITY AND DIGITAL FORENSICS ENGINEERING

COURSE CODE: IA 317

GROUP NUMBER 02:

GROUP ASSIGNMENT

JAMES ALBERT MWINGIRA	T23-03-10342
LIGHTNESS NIXON TUMAINI	T23-03-02762
HELENA MAYOMBE	T23-03-12761
JACKSON JOHN SHIGELA	T23-03-09266
RUUVULY ABDI SELEMANI	T23-03-16835

Tracking Medical Supply Donations Using Blockchain

Problem Statement:

Donation procedures for medical supplies frequently suffer from a lack of accountability, traceability, and transparency. Poor record-keeping, manual tracking, or corruption can cause donated items to be misallocated, lost, or delayed. The flow of supplies from donors to distribution centers and ultimately to healthcare facilities is often not visible to beneficiaries or donors. This can lead to shortages or unequal distribution of vital medical resources, as well as a decrease in trust and an increase in inefficiencies.

To guarantee safe, impenetrable tracking and real-time visibility of medical donations across the whole supply chain, a solution is required.

Main Objective:

To design and implement a blockchain-based system that ensures secure, transparent, and tamper-proof tracking of medical supply donations from donors to final recipients.

Specific Objective:

1. To develop a decentralized ledger for recording all donation transactions in an immutable format.
2. To enable real-time tracking of medical supplies across the donation and distribution chain.
3. To create a user interface for donors, NGOs, and healthcare facilities to register, verify, and update donation data.

4. To automate verification processes using smart contracts to prevent fraud and unauthorized modifications.
5. To generate audit trails and reports that enhance accountability and trust among stakeholders.
6. To evaluate system performance in terms of transparency, security, and ease of use.